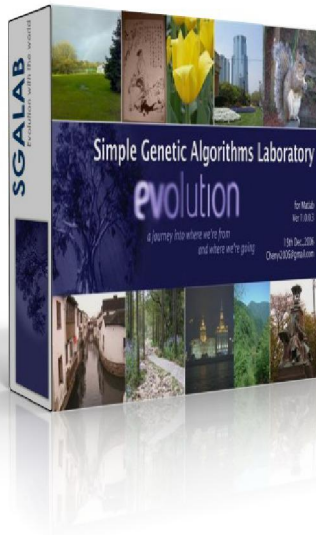




SGALAB

Evolution with the world

SxLAB Family Member



SGALAB Quick Start

(3) Micro-GA (muGA, μ GA)

and

GA with Mendel's Principles



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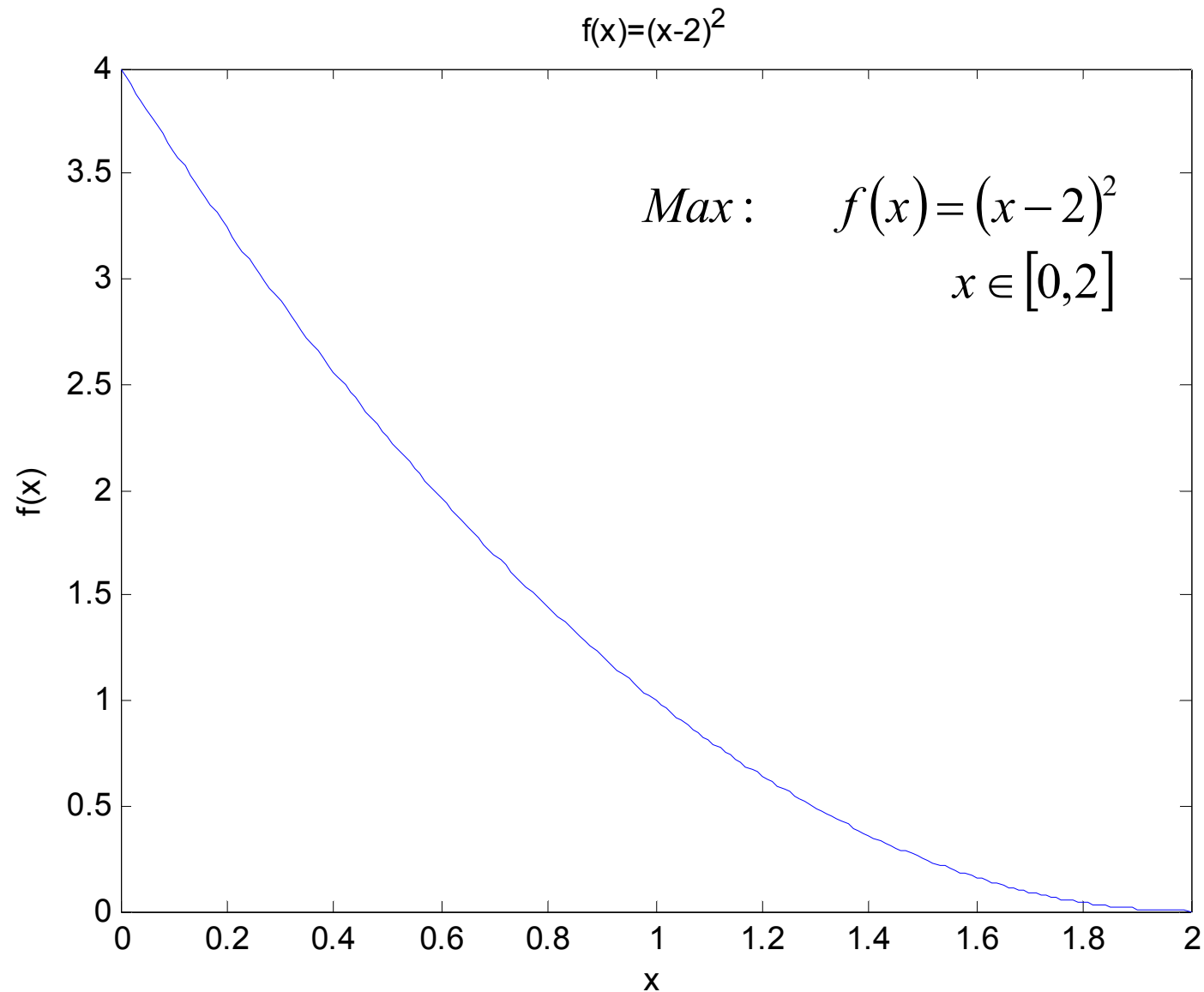
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Case Description – Single Objective Without Constraints



Step 1 – Define the single Objective in fitness function

SGA_FITNESS_function.m

```
function [ fitness ] = SGA_FITNESS_function( x )
```

```
%SGA_FITNESS_function begin
```

```
%User can design their own fitness function here
```

```
%as a standard matlab function
```

```
fitness = (x-2)^2;
```

```
%SGA_FITNESS_function end
```

Step 2 – Set parameters

Parameters	Value	file
<i>Max Generation</i>	<i>100</i>	<i>INPUT_max_generation.txt</i>
<i>Crossover Probability</i>	<i>0.8</i>	<i>INPUT_probability_crossover.txt</i>
<i>No mutation operation</i>		
<i>Population</i>	<i>30</i>	<i>INPUT_population.txt</i>
<i>step</i>	<i>0.01</i>	<i>INPUT_steps.txt</i>
<i>Convergence method</i>	<i>1</i>	<i>INPUT_convergence_method.txt</i>
<i>max confines</i>	<i>2</i>	<i>INPUT_max_confines.txt</i>
<i>min confines</i>	<i>0</i>	<i>INPUT_min_confines.txt</i>
<i>No constraints</i>	<i>0</i>	<i>Options(5) = 0 in SGALAB_demo_single_objective_muGA.m</i>

Step 2 – Set parameters for Micro-GA(muGA)

Parameters	Value	file
<i>Micro-GA cycle (internal loop)</i>	4	<i>INPUT_muGA_cycle.txt</i>
<i>Micro-GA population *</i> <i>(internal loop)</i>	6	<i>INPUT_muGA_population_internal_number.txt</i>
<i>Replaceable population</i>	2	<i>INPUT_muGA_population_replaceable_number</i>
<i>Exchangeable population **</i>	2	<i>INPUT_muGA_exchangeable_number.txt</i>

* *Micro-GA population* = *Replaceable population* + *Non-replaceable population*

** *Data exchange between Micro-GA population with whole population*

Step 3 – Solve and results

Run '[SGALAB_demo_SO_muGA](#)'
in matlab command window

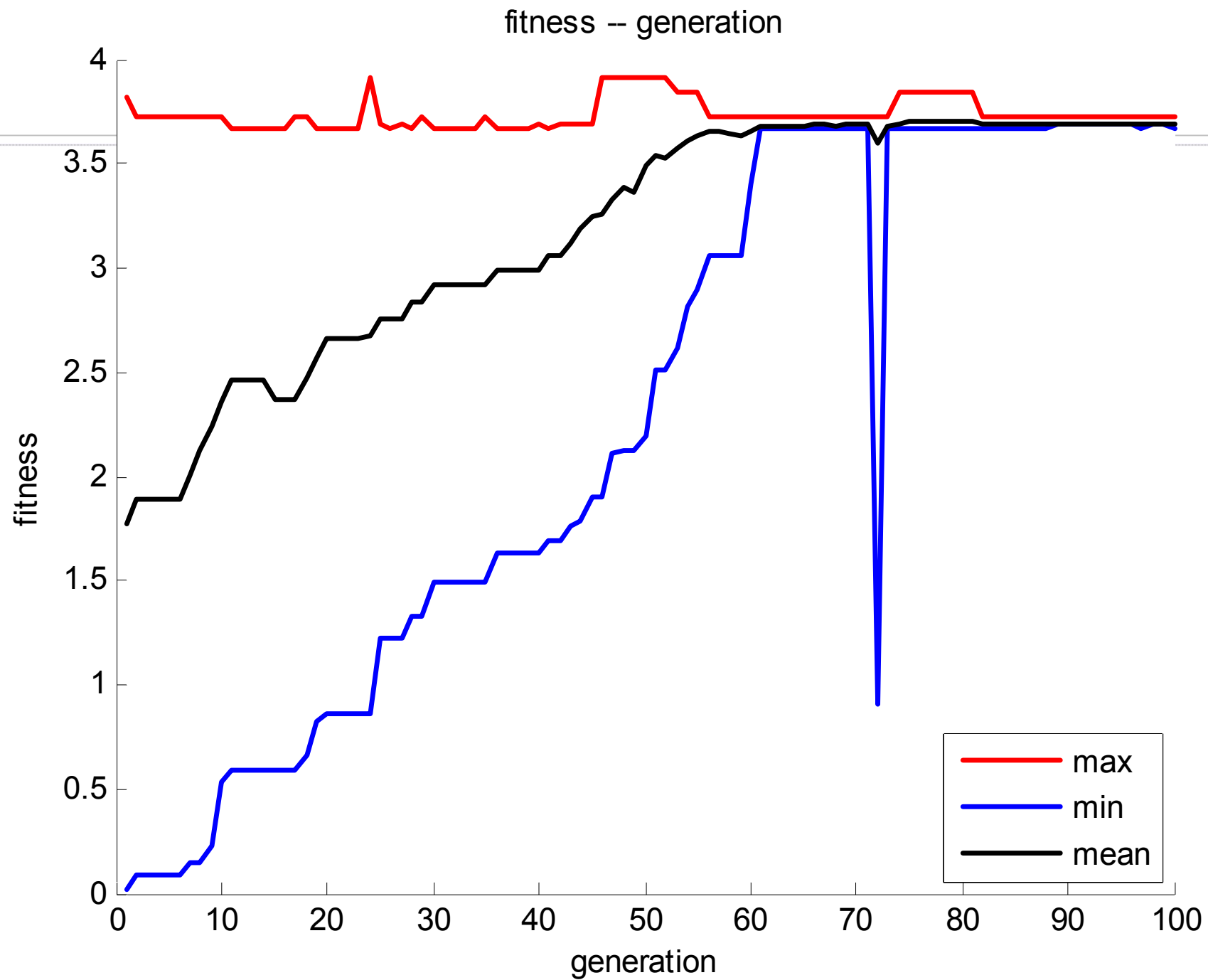
% set in [SGALAB_demo_SO_muGA.m](#) options

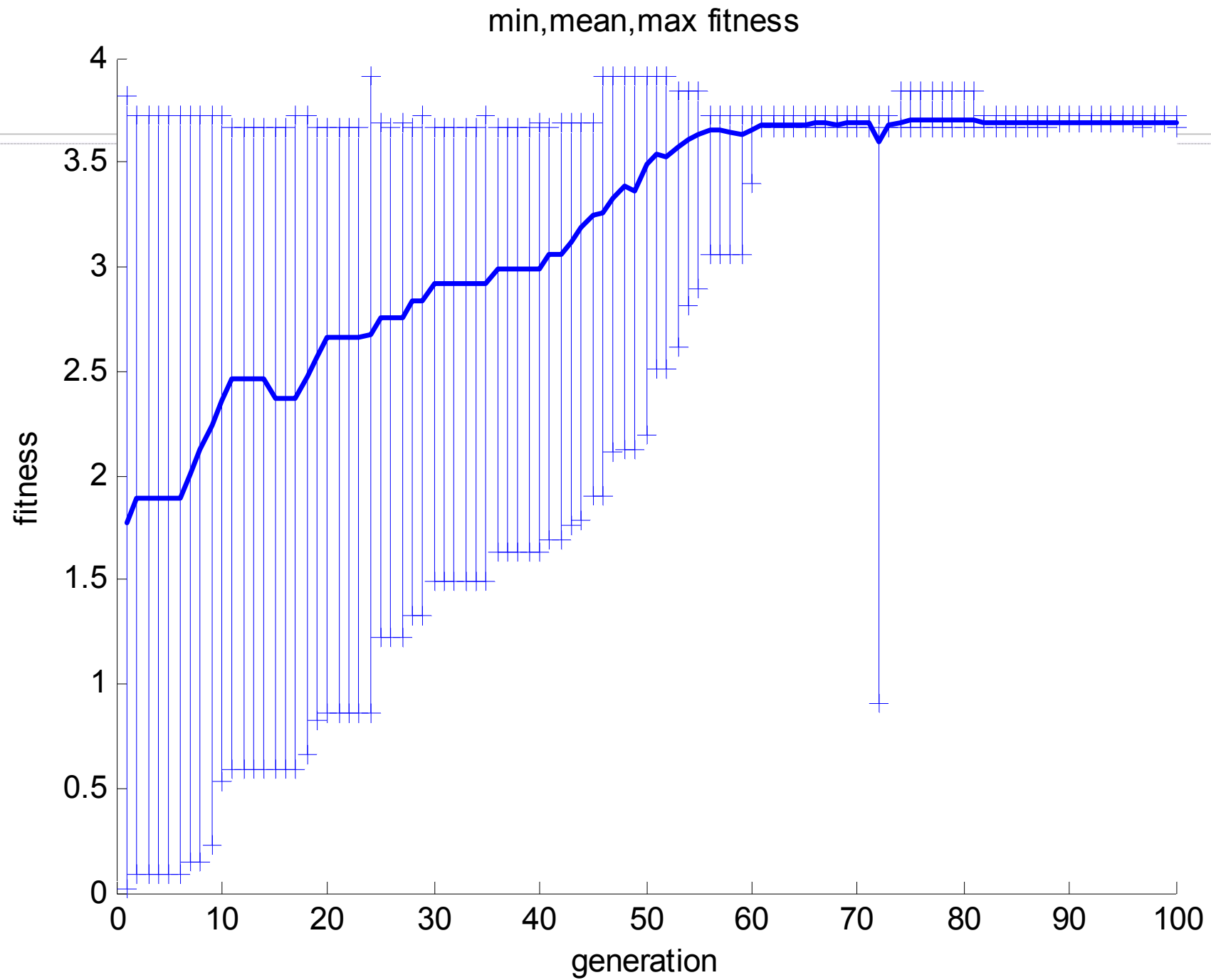
```
options = { 'Binary',  
            'Roulettewheel',  
            'singlepoint',  
            'singlepoint',  
            '0',  
            'NON_MO', % will check inside SGA_entry_MO_Pareto_muGA()  
            '0', % ignore  
            '0', % ignore  
            '1'}; % Micro-GA option: 1 - invoke mutation operator  
            % 0 - NOT invoke mutation operator
```

Step 3 – Solve and results

Results in 'OUTPUT_*.txt'

Results	Value	file
$F(x_0)_{\max}$	3.90	OUTPUT_maxfitness.txt
$F(x_0)_{\min}$	0.013	OUTPUT_minfitness.txt
$F(x_0)_{\text{mean}}$	3.176	OUTPUT_meanfitness.txt
x_0	0.078	OUTPUT_best_result_space.txt
Cost time (sec.)	1.11	In command window





If more complex fitness function

$$\text{Max} : f(x, y, z) = x \sin(10\pi x + y + z) + 2.0$$

$$-1 \leq x \leq 2$$

$$-10 \leq y \leq 50$$

$$10 \leq z \leq 20$$

```
function [ fitness ] = SGA_FITNESS_function( x,y,z )
```

```
%SGA_FITNESS_function begin
```

```
%User can design their own fitness function here
```

```
%as a standard matlab function
```

```
fitness = x*sin(10*pi*x+y+z)+2.0;
```

```
%SGA_FITNESS_function end
```

If more complex fitness function

INPUT_max_confines.txt

2

50

20

INPUT_min_confines.txt

-1

-10

10

INPUT_steps.txt

0.01

0.01

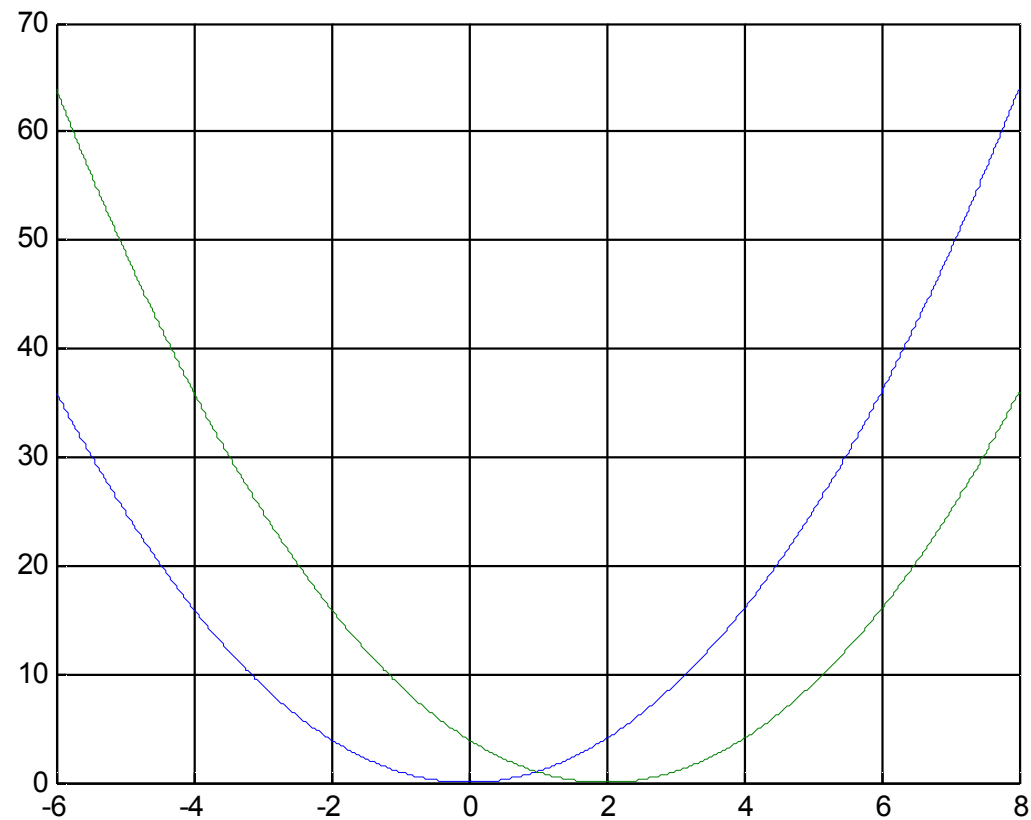
0.01

Contents

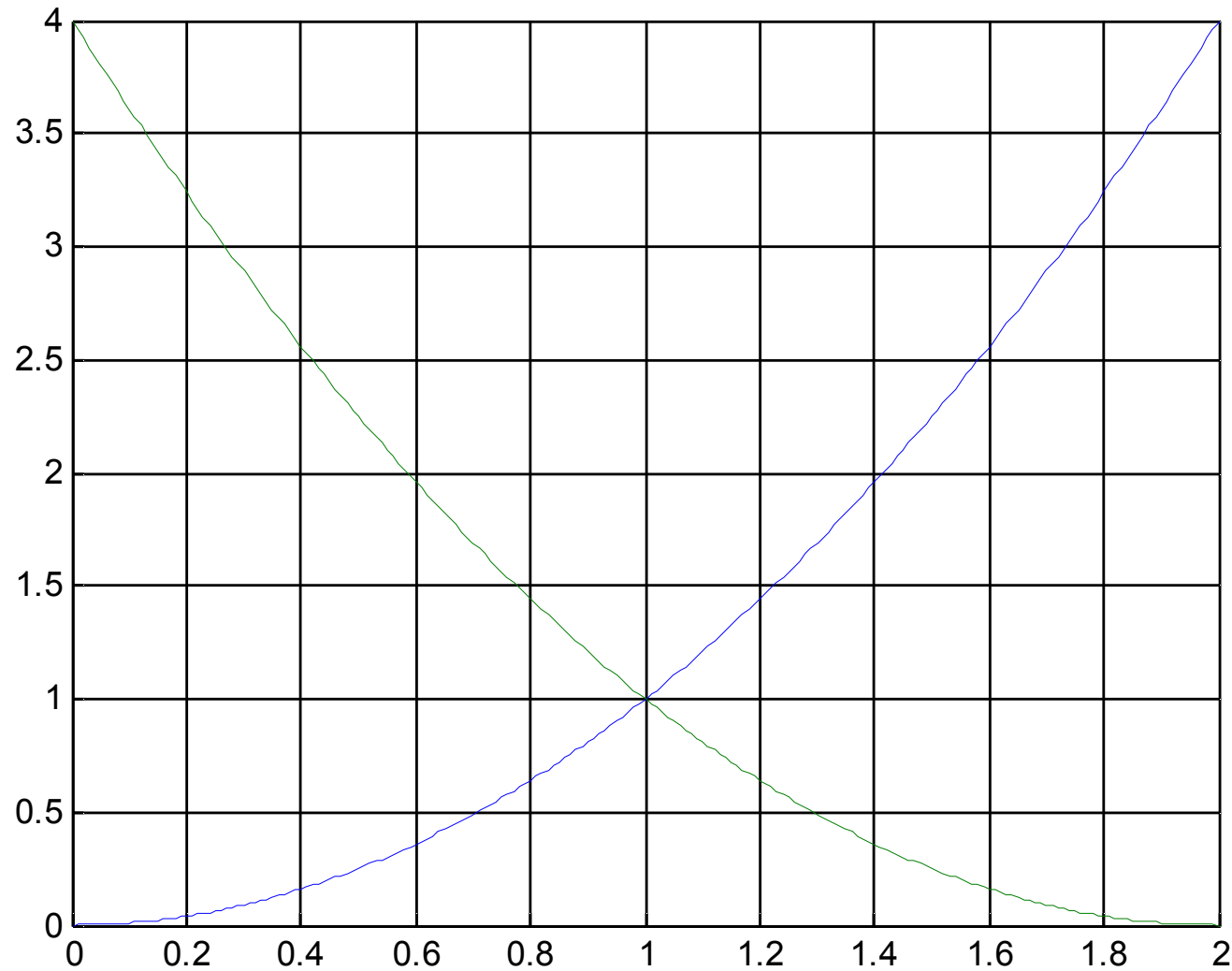
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Case Description – Multi-Objective With Constraints

$$\begin{cases} f_1(x) = x^2 \\ f_2(x) = (x-2)^2 \end{cases} \quad x \in [0, 2]$$



Case Description – Multi-Objective Problem – Graph Zoom in



Step 1 – Define the single Objective in fitness function

SGA_FITNESS_function.m

$$\begin{cases} f_1(x) = x^2 \\ f_2(x) = (x-2)^2 \end{cases} \quad x \in [0,2]$$

```
function [ MO_fitness_array ] = SGA_FITNESS_function( x )
```

```
%SGA_FITNESS_function begin
```

```
%User can design their own fitness function here
```

```
%as a standard matlab function
```

```
...
```

```
%User Define - fitness function 1 :
```

```
fitness1 = x^2;
```

```
%
```

```
%User Define - fitness function 1 :
```

```
fitness2 = (x-2)^2;
```

```
%set output
```

```
MO_fitness_array = [ ...
```

```
    fitness1...
```

```
    fitness2 ];
```

```
...
```

```
%SGA_FITNESS_function end
```

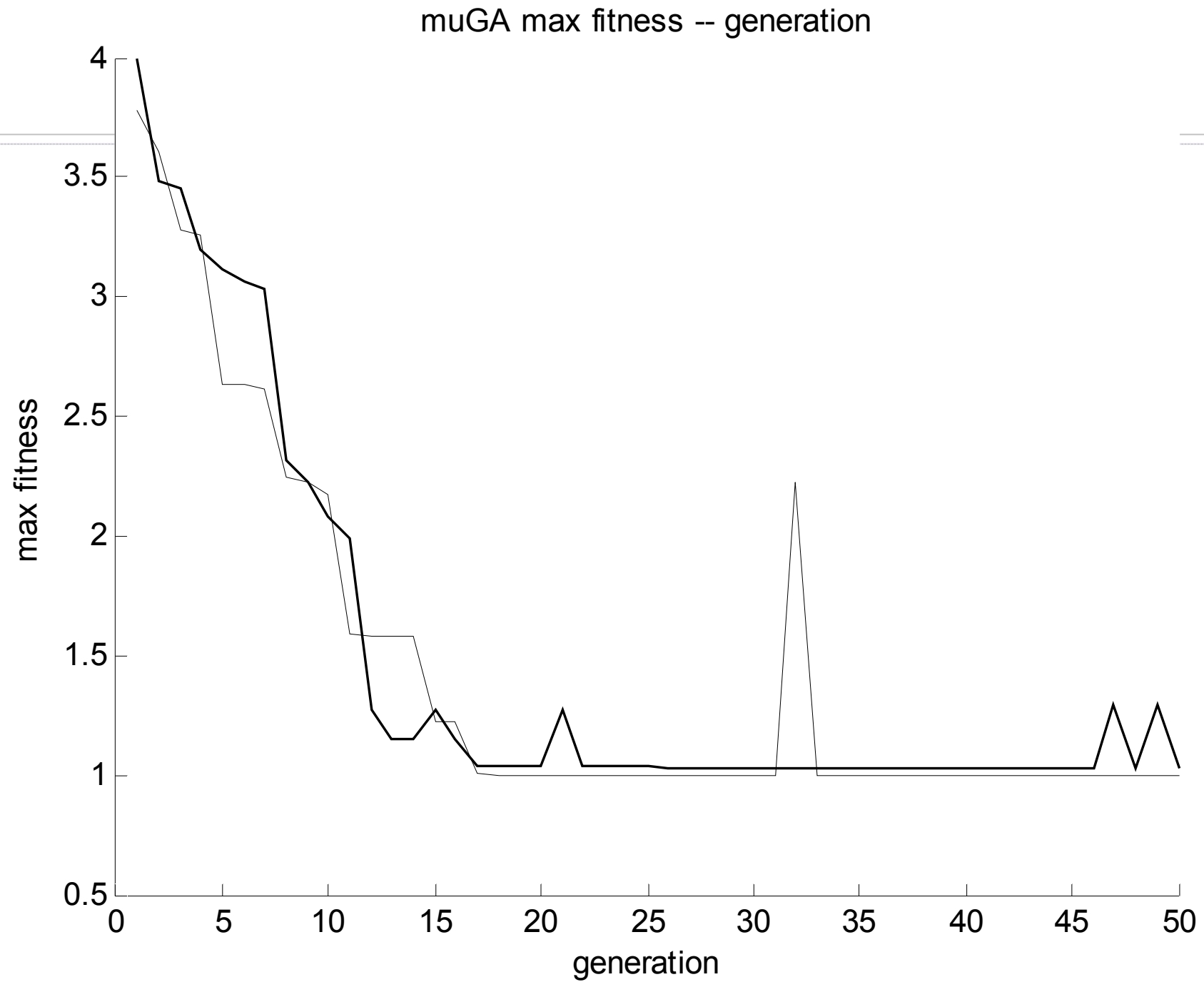
© SSALAB : 2016. All rights reserved.

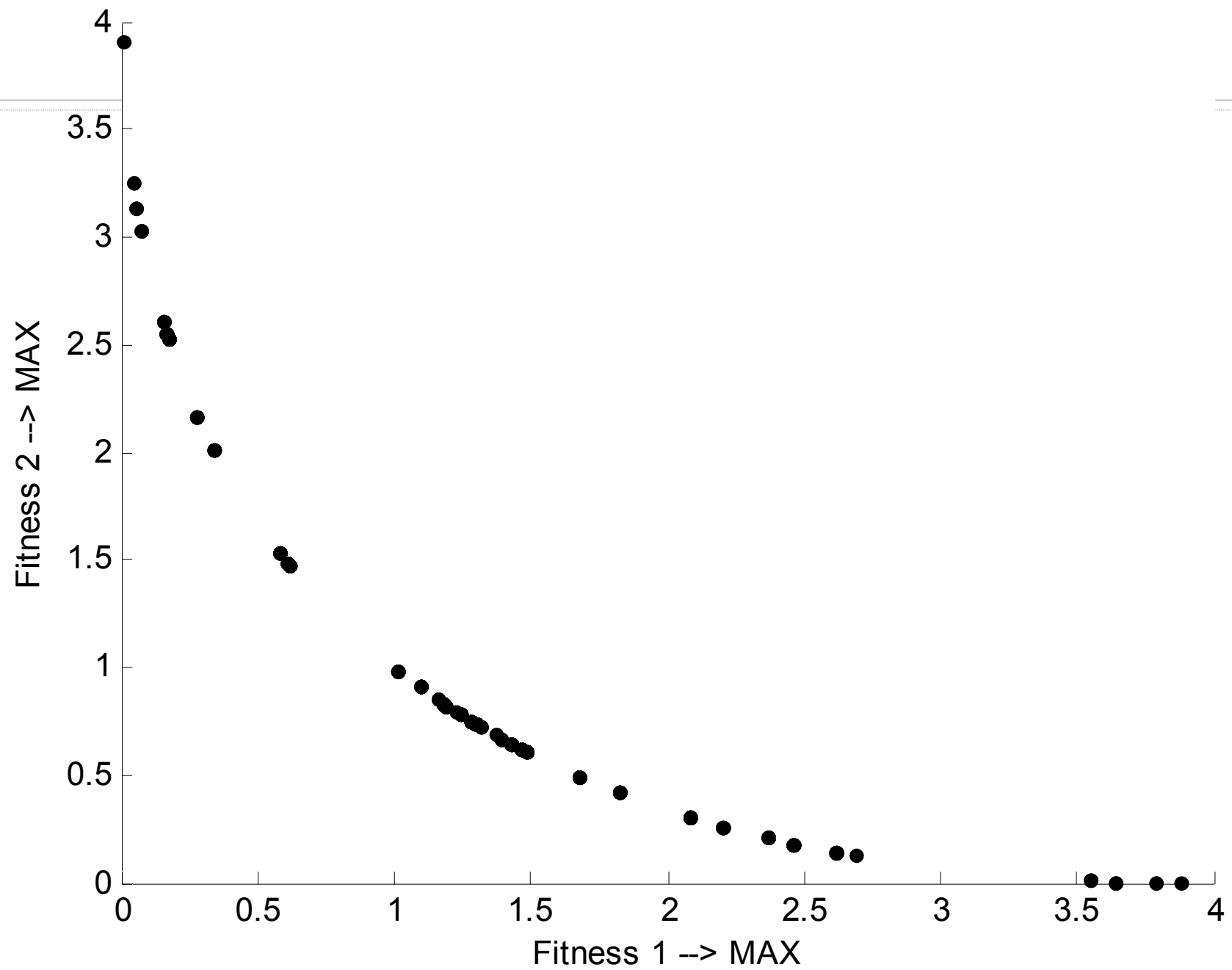
Step 2 – Solve and results

Run '[SGALAB_demo_MO_muGA](#)' in matlab command window

% set in [SGALAB_demo_MO_muGA.m](#) options

```
options = { 'Binary',  
            'Roulettewheel',  
            'singlepoint',  
            'singlepoint',  
            '0',  
            'muga', % will check inside SGA_entry_MO_Pareto_muGA()  
            'Goldberg',  
            '1',  
            '1'}; % Micro-GA option:  
            %      1 - invoke mutation operator  
            %      0 - NOT invoke mutation operator
```



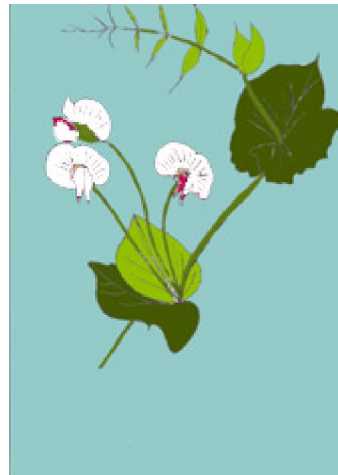
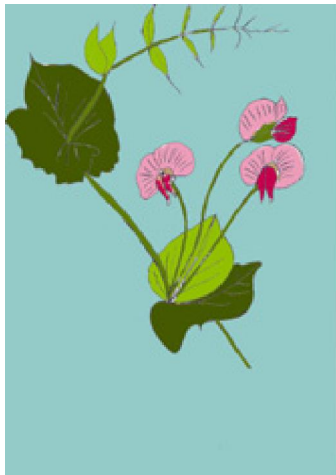


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Case Description – Single Objective Without Constraints

Same as [Case Study 6](#)



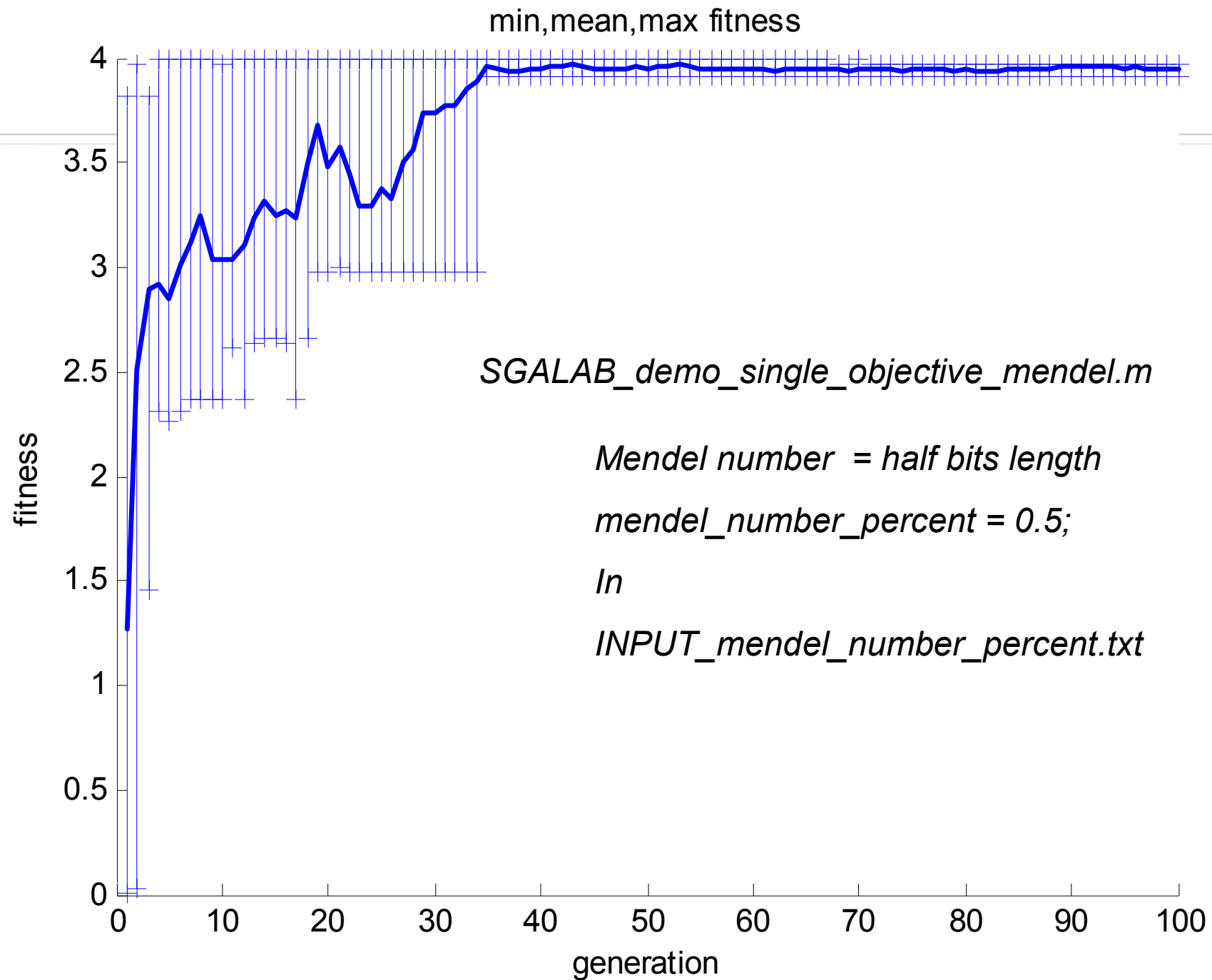
Step 2 – Set parameters for Mendel's Law Operator)

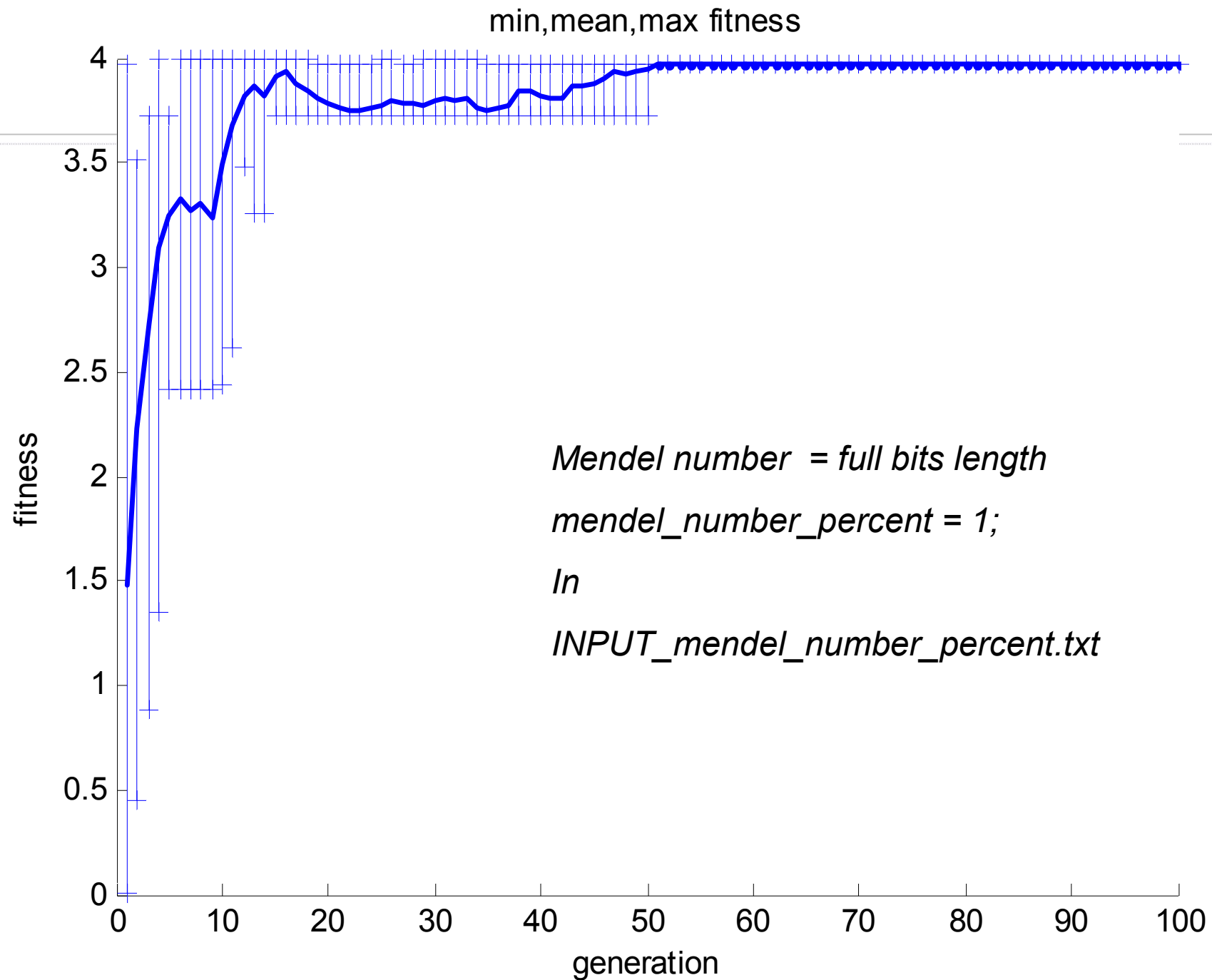
Parameters	Value	file
<i>Mendel Number in percentage</i>	$MN \in (0, 1]$ <i>In default, 1</i>	<i>INPUT_mendel_number_percent.txt</i>

0D	1D	0D	1R	0R	1R
----	----	----	----	----	----

←————→ *Mendel Number = bits length*

←————→



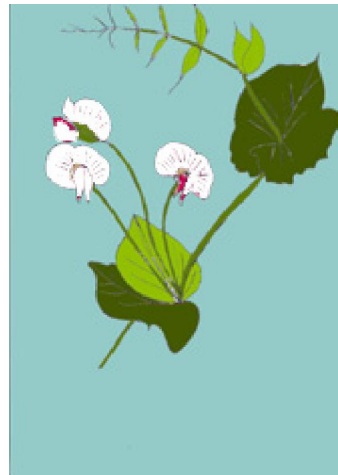
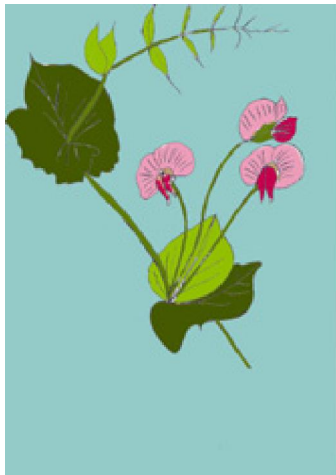


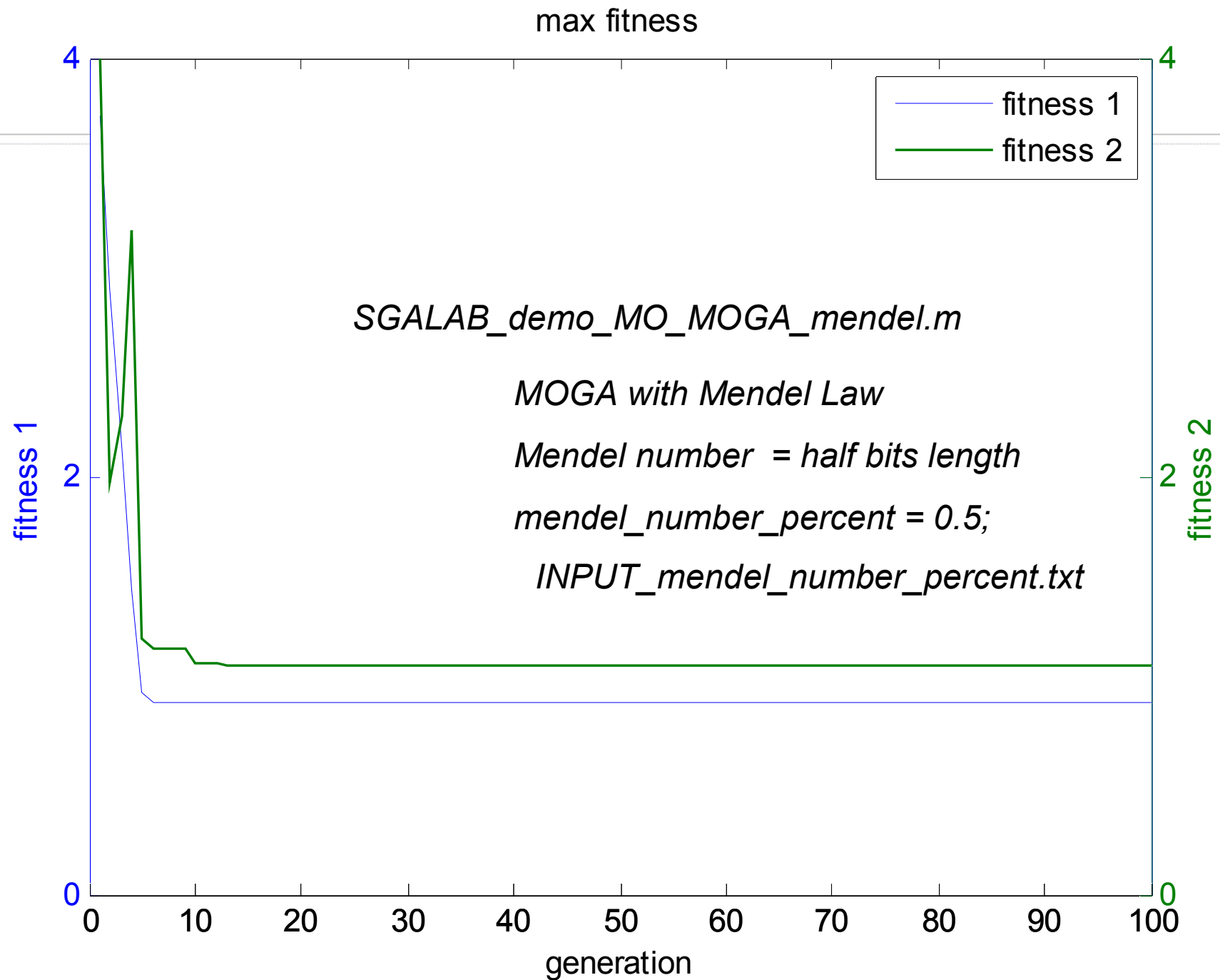
Contents

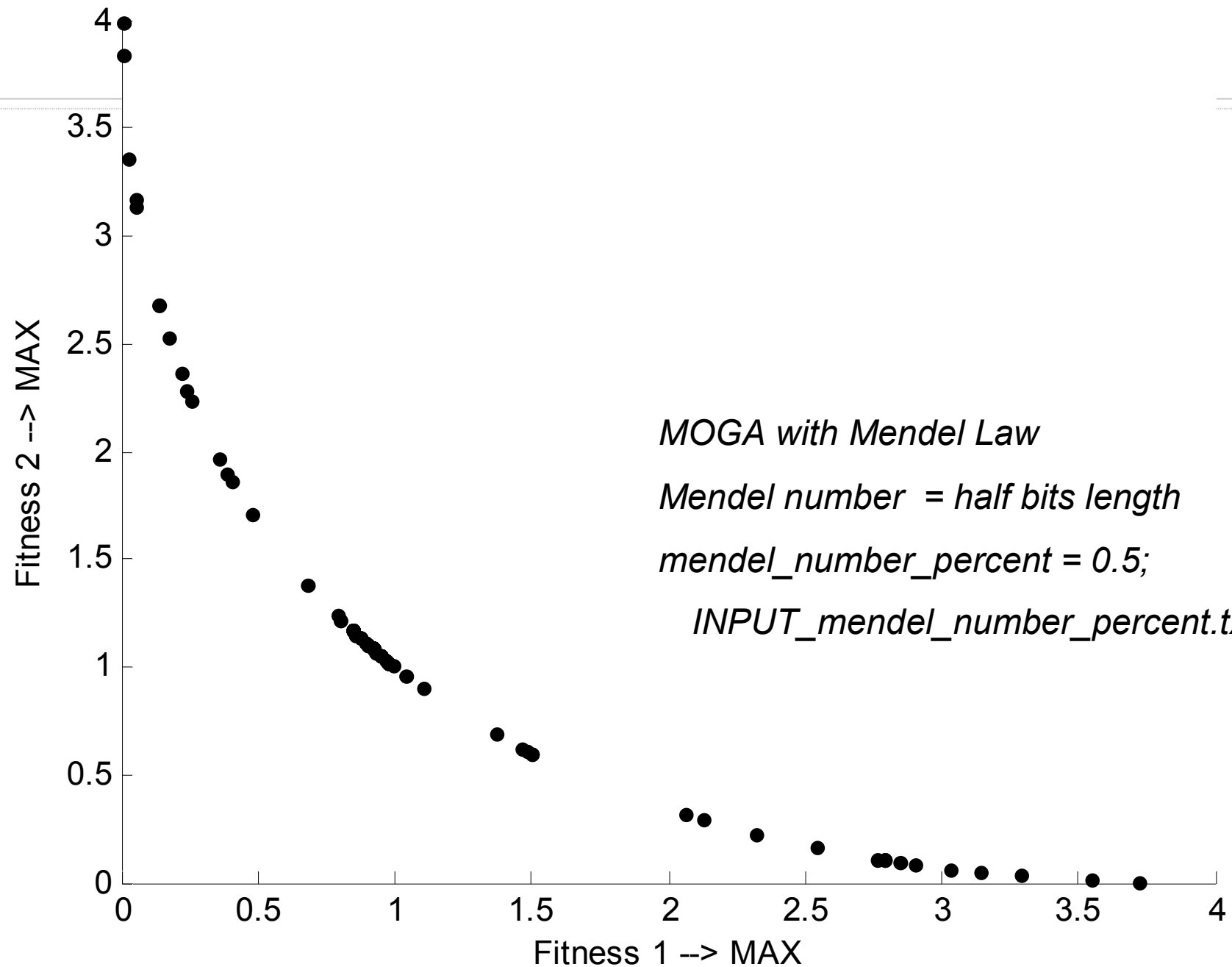
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Case Description – Multi-Objective Without Constraints

Same as [Case Study 7](#)







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FAQ 1 - Gregor Mendel

His hybridisation studies with pea plants that “the behaviour of each pair of differentiating characteristics *[e.g. the shape and colour of seeds]* in hybrid union is independent of the other differences between the two original plants” (Mendel 1866).

This formulation was actually the only regularity that Mendel referred to as a “law”.

In Carl Correns’ 1900 paper (one of a trio appearing that year which are considered to represent the rediscovery of Mendel) he refers to this as Mendel’s Law (Correns 1900; Olby 1966).

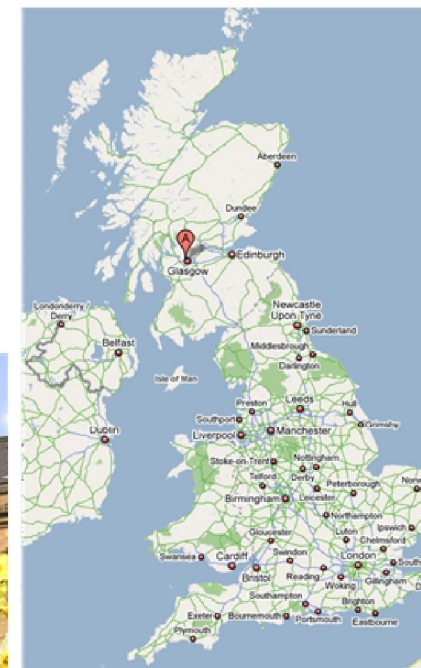
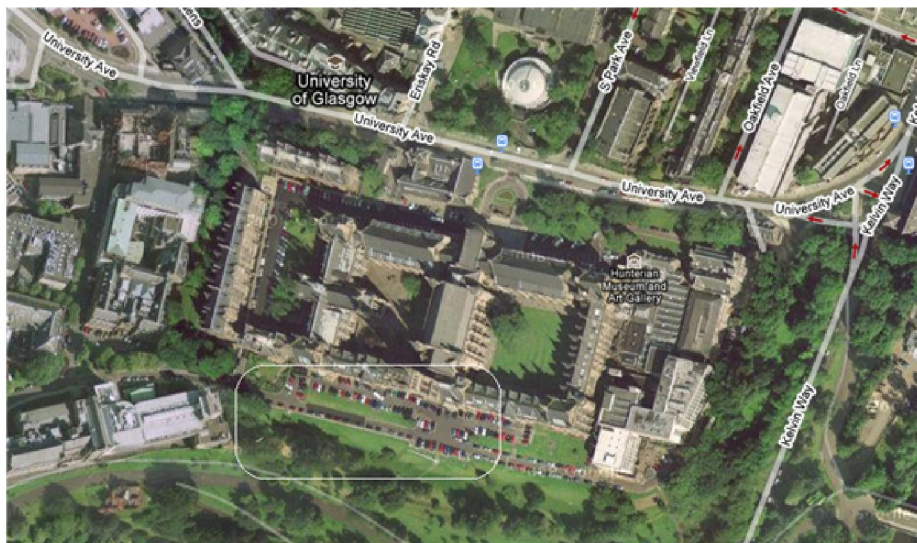
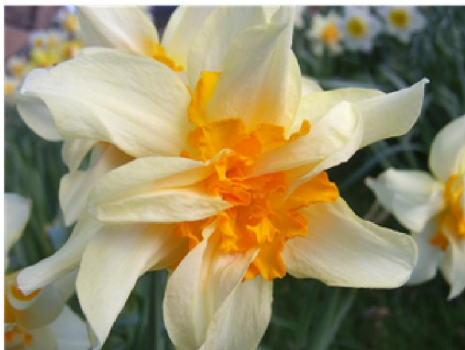
[1] http://en.wikipedia.org/wiki/File:Gregor_Mendel_Monk.jpg

[2] <http://www.bbc.co.uk/dna/h2g2/A885521>

Gregor Mendel (1822-1884)



Daffodil



ADS Color Classification

W = White or whitish

G = Green

Y = Yellow

P = Pink

O = Orange

R = Red

Any Questions?

→ See 'SGALAB User's Guide' and other documents

→ Send email to leo.chen.yi@gmail.com ,
but no sure reply in time, sorry about that.^_^



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雨霖铃

古楼孤灯，雪褪残寒，晓径蔓延。朝烟散绒芽翠，短短桃红，轻轻柳絮。渐渐飞舞无数，临水沾人衣。翩翩樱樱信风天，青霜白草月华见。

寒塘冷雾花对影，暖红袖添香夜读书。世间路多艰难，经历处，笔头千言。往事千端，总有十年晴朗婵娟。人生若只如初见，斑斑点点。

End

